

CS & IT ENGINEERING

Discrete Mathematics

Mathematical Logic

DPP-02 Discussion Notes



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#Q. Let $p(x)$ and $q(x)$ denote the following open statements.

✓ $p(x): \underline{x^2 > 0}$

✓ $q(x): \underline{x \text{ is odd}}$

for the universe of all integers, determine the truth or falsity of each of the statement.

✗ $S_1: \forall x [p(x) \rightarrow q(x)]$

✓ $2^2 = 4 > 0$
 ↓
 no

$3^2 = 9 > 0$
 ↓
 odd

✓ $S_2: \exists x [p(x) \rightarrow q(x)]$ ✓

which of the following is true?

A S_1 only

✓ **B** S_2 only

C Both S_1 and S_2

D Neither S_1 nor S_2

#Q. Consider following two First Order Logic Statements:

$$S_1: [\forall x(\sim P(x) \vee Q(x))] \rightarrow [\forall x P(x)] \rightarrow [\forall x Q(x)]$$

$$S_2: [\exists x P(x)] \rightarrow [\exists x Q(x)] \rightarrow [\exists x (P(x) \rightarrow Q(x))]$$

Handwritten notes for S_1 : $P_1 \wedge P_2$ above the second $\rightarrow$; $(P_1 \vee P_2) \rightarrow (Q_1 \vee Q_2)$ in a box; $\rightarrow ((P_1 \rightarrow Q_1) \vee (P_2 \rightarrow Q_2))$ below the box.

Which of the following is valid?

A S_1 only

B S_2 only

C Both S_1 and S_2

D None of these

$$(\bar{P}_1 \wedge \bar{P}_2) \vee Q_1 \vee Q_2 \rightarrow \{ \bar{P}_1 \vee Q_1 \vee \bar{P}_2 \vee Q_2 \}$$

$\equiv 1$

#Q. $P(y) = \sqrt{y}$ is real in the domain of $\mathbb{Z}^+$, then which of the following is / are correct?

☒ **A**

$\forall y P(y)$

True

☒ **B**

$\exists y P(y)$

$\sqrt{y}$

two integers

☒ **C**

$\forall y \sim P(y)$

☒ **D**

$\exists y \sim P(y)$

#Q. Which of the following is not valid logical expression?

A

$$\forall x [P(x) \rightarrow Q(x)] \rightarrow [\forall x P(x)] \rightarrow [\forall x Q(x)]$$

$$[(P_1 \vee Q_1) \wedge (P_2 \vee Q_2)]$$

$$\rightarrow ((P_1 \wedge P_2) \vee (Q_1 \wedge Q_2))$$

B

$$\forall x [P(x) \vee Q(x)] \rightarrow [\forall x P(x)] \vee [\forall x Q(x)]$$

$$(P_1 P_2 + P_1 Q_2 + P_2 Q_1 + Q_1 Q_2)$$

C

$$\exists x [P(x) \wedge Q(x)] \rightarrow [\exists x P(x)] \wedge [\exists x Q(x)]$$

$$\rightarrow (P_1 P_2 + Q_1 Q_2)$$

D

$$\forall x [P(x) \leftrightarrow Q(x)] \rightarrow [\forall x P(x)] \leftrightarrow [\forall x Q(x)]$$

$$\equiv P_1 P_2 + Q_1 Q_2 + P_1' + Q_1' + P_2' + Q_2'$$

[MCQ]



#Q. Consider following logical expressions:

I: $\forall y[P(y) \rightarrow Q] \leftrightarrow [\forall y P(y)] \rightarrow Q$

II: $\exists y[P(y) \rightarrow Q] \rightarrow [\exists y P(y)] \rightarrow Q$ —

which of the following logical expression is valid?

A

I only

$$\left\{ \left[(\bar{p}_1 + Q) (\bar{p}_2 + Q) \right] \leftrightarrow p_1 p_2 \right\} \rightarrow Q$$

B

II only

$$\left\{ (\bar{p}_1 \bar{p}_2 + Q) \leftrightarrow p_1 p_2 \right\} \rightarrow Q$$

C

Both I and II

$$\left\{ (\bar{p}_1 \bar{p}_2 + Q) p_1 p_2 + (\bar{p}_1 \bar{p}_2 + Q) \overline{p_1 p_2} \right\} \rightarrow Q$$

D

None of these

$$(p_1 p_2 Q + \bar{p}_1 \bar{p}_2 + \bar{p}_1 \bar{p}_2 Q) \rightarrow Q$$

[MCQ]



#Q. Consider
Actor (x) = x is an actor
Smart (x) = x is smart
and the well-formed formula:

$$\exists x (\text{Actor}(x) \wedge \text{Smart}(x))$$

Choose the correct representation of above in English sentence.

$$(p_1 p_2 Q + p_1 \overline{p_2}) \rightarrow Q$$

$$(\overline{p_1} + \overline{p_2} + \overline{Q}) (\overline{p_1} + p_2) + Q$$

$$\overline{p_1} p_2 + p_1 \overline{p_2} + p_1 \overline{Q} + p_2 \overline{Q} + Q$$

☒ **A**

Some Actor is smart.

☐ **B**

Some Actor is not smart.

☐ **C**

All actors are smart.

☐ **D**

All smart are actors.

#Q. Consider the following statement

✓ “There is exactly one apple”. ✓

Let $G(x)$: x is an apple.

Now consider the predicate logic statements:

✓ I. $\exists x \text{ apple}(x) \wedge \forall y (\text{apple}(y) \Rightarrow x = y)$

✗ II. $\exists x \text{ apple}(x)$ $\Rightarrow$ True — at least one apple

The correct representation in predicate logic is ?

✓ **A** Only I

B Only II

C Both I and II

D Neither I and II

#Q. Consider the following statements:

I. $\exists x \{p(x) \rightarrow \{\forall x P(x) \rightarrow \exists x Q(x)\}\}$

II. $\exists x \forall y P(x, y) \rightarrow \forall y \exists x P(x, y)$

The number of valid statements is/are 2.

#Q. Choose among the following that are not equivalent to the given first order logic statement:

$$\underline{(\exists x) (\forall y) [p(x, y) \wedge q(x, y) \wedge \neg r(x, y)]}$$

$$p \quad q \quad \bar{r}$$

☒ **A** $(\forall x) (\exists y) [p(x, y) \wedge q(x, y) \rightarrow r(x, y)]$

☒ **B** $(\exists x) (\forall y) [p(x, y) \vee q(x, y) \wedge \neg r(x, y)]$

☒ **C** $\neg (\forall x) (\exists y) [p(x, y) \vee q(x, y) \rightarrow r(x, y)]$

☒ **D** $\neg (\forall x) (\exists y) [p(x, y) \wedge q(x, y) \rightarrow r(x, y)]$

$$\begin{aligned} & (p \wedge q) \rightarrow r \\ & (\bar{p} + \bar{q} + r) \end{aligned}$$

#Q. Choose the correct representation for the below statement:

Player(x): x is a player

Coach(y): y is a coach

Likes (y, x): y likes x

“Every player is liked by some coach”

☒ **A** $\forall (x) [\text{player}(x) \rightarrow \exists y [\text{coach}(y) \wedge \text{likes}(y, x)]]$

☐ **B** $\forall (x) [\text{player}(x) \rightarrow \exists y [\text{coach}(y) \rightarrow \text{likes}(y, x)]]$

☐ **C** $\exists (x) [\text{player}(x) \rightarrow \forall y [\text{coach}(y) \rightarrow \text{likes}(y, x)]]$

☐ **D** $\exists (x) [\text{player}(x) \rightarrow \forall y [\text{coach}(y) \wedge \text{likes}(y, x)]]$



THANK - YOU